

Zhenyu (Fischer) Lei

CONTACT INFORMATION

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*Ph.D. student (expected Dec 2027) working on LLM reasoning, post-training, agents, memory, and self-improvement. Seeking **Summer 2027 research internships**.*

RESEARCH INTERESTS

- **LLM Reasoning & Post-training:** Distillation; Reinforcement Learning (GRPO/RLVR);
- **LLM Agents & Memory:** Agent Memory; Recursive Self-Improvement; Agent Harnesses
- **Model Editing & Interpretability:** Reasoning Editing; Mechanistic Interpretability
- **Time Series:** Time Series & Spatio-temporal Forecasting; Few-shot / Inductive Forecasting

INDUSTRIAL EXPERIENCES

Capital One PhD Fellow, Capital One 2026 – Present
• Researching cost-efficient verification for recursively self-improving LLM agent harnesses.
• Collaborating with Capital One research scientists on agentic LLM systems.

Applied Scientist Intern, AT&T Labs Research, Bedminster, NJ Jun 2025 – Aug 2025
• Improved LLM reasoning through model editing guided by mechanistic interpretability.
• Resulting work published at ICLR 2026 [C4]. Host: Dr. Qiong Wu.

EDUCATION

University of Virginia, Charlottesville, VA, USA

Ph.D., Electrical and Computer Engineering Aug 2023 – Dec 2027 (expected)
• Advisor: Dr. Jundong Li
• GPA: 3.89/4.0

Xi'an Jiaotong University, Xi'an, China

B.S., Physics (Honor Graduation) Aug 2019 – Jul 2023

PROJECTS

Good Papers (goodpapers.org) *Creator & Developer*, 2026
• Public platform for rating research papers with human and AI (LLM persona) ratings; API for LLM agents to query and review papers.

PyGDebias *Main Contributor*
• Open-source library for fairness-aware graph learning: 13 debiasing methods, 26 graph datasets, unified benchmarks (WWW Companion 2024).

MoCo: Model Collaboration *Contributor*
• One-stop toolkit of methods and benchmarks for multi-LLM collaboration research.

TECHNICAL SKILLS

- **LLM Post-training:** SFT, RL for LLMs (GRPO, PPO, DPO, RLVR), Knowledge/Reasoning Distillation, LoRA/QLoRA & PEFT, Chain-of-Thought & Test-time Scaling
- **LLM Agents:** Agent Memory, Tool Use, Agent Harnesses & Scaffolds, Self-improving Agents, RAG, LLM-as-a-Judge, Agent Evaluation
- **Editing & Interpretability:** Knowledge Editing (ROME, MEMIT, etc.), Circuit Discovery, Activation Patching, Probing
- **Multimodal:** Vision/Molecule-Language Models, Multimodal Knowledge Editing, Multimodal Fusion (fMRI/DTI), Multimodal Property Prediction

- **Time Series & Graphs:** Time Series & Spatio-temporal Forecasting, Graph Neural Networks, Graph Prompting, Fairness & Unlearning
- **Frameworks:** PyTorch, Hugging Face (Transformers, PEFT, TRL, Accelerate, Datasets), verl, OpenRLHF, vLLM, SGLang, DeepSpeed, FSDP, PyTorch Geometric, LangChain/LangGraph
- **Models:** Llama, Qwen, Mistral, DeepSeek(-R1-Distill), GPT & Claude APIs
- **Infrastructure:** Multi-GPU/Multi-node Training, Slurm, Docker, Linux, Git, Weights & Biases
- **Languages:** Python, C/C++, SQL, JavaScript/HTML, $\LaTeX$

HONORS AND AWARDS	• Capital One PhD Fellowship	2026
	• WSDM Student Travel Award	2026
	• UVA Thriving Youth in a Digital Environment (TYDE) Student Training Award	2026
	• UVA Presidential Fellowship in Collaborative Neuroscience (One of <i>six</i> recipients university-wide)	2025
	• UVA Precision Health Initiative Trainee Travel Stipend	2025
	• UVA Engineering Academic Commitment to Excellence (ACE) Fellowship	2024

CITATION STATISTICS [Google Scholar](#) (as of Oct 2026): Citations **734**, h-index **11**, i10-index **11**

PUBLICATIONS • **23** Conference Papers, **2** Journal Papers, **1** Tutorial. * denotes equal contribution.

Conference Publications

- [C1] **Zhenyu Lei**, Chenghao Liu, Yushun Dong, Qi R. Wang, Jundong Li. “AdaST: Adaptive Coupling for Spatial-Temporal Forecasting”, Annual Conference on Neural Information Processing Systems (**NeurIPS**), 2026.
- [C2] **Zhenyu Lei**, Mingjia Shi, Xingbo Fu, Haoyu He, Qi R. Wang, Jundong Li. “Grounding Memory Summarization in Utility Intent”, Conference on Empirical Methods in Natural Language Processing (**EMNLP**), 2026. (**Oral Presentation**)
- [C3] **Zhenyu Lei**, Zihan Chen, Yaochen Zhu, Shangbin Feng, Zaiyi Zheng, Ruocheng Guo, Yushun Dong, Jundong Li. “TeacherGRPO: Closing the Capacity Gap in Reasoning Distillation via Teacher Alignment”, Findings of EMNLP (**EMNLP Findings**), 2026.
- [C4] **Zhenyu Lei**, Qiong Wu, Jianxiong Dong, Yinhan He, Emily Dodwell, Yushun Dong, Jundong Li. “Reforming the Mechanism: Editing Reasoning Patterns in LLMs with Circuit Reshaping”, International Conference on Learning Representations (**ICLR**), 2026. [Paper]
- [C5] **Zhenyu Lei**, Patrick Soga, Yaochen Zhu, Yinhan He, Yushun Dong, Jundong Li. “MolEdit: Knowledge Editing for Multimodal Molecule Language Models”, ACM International Conference on Web Search and Data Mining (**WSDM**), 2026. (**Oral Presentation**) [Paper]
- [C6] **Zhenyu Lei**, Aiyang Zhang, Song Wang, Han Fan, Jundong Li. “BrainTAP: Brain Disorder Prediction with Adaptive Distill and Selective Prior Integration”, IEEE International Symposium on Biomedical Imaging (**ISBI**), 2026.
- [C7] **Zhenyu Lei**, Zhen Tan, Song Wang, Yaochen Zhu, Zihan Chen, Yushun Dong, Jundong Li. “Learning from Diverse Reasoning Paths with Routing and Collaboration”, Conference on Empirical Methods in Natural Language Processing (**EMNLP**), 2025. [Paper]

- [C8] **Zhenyu Lei**, Yushun Dong, Weiyu Li, Rong Ding, Qi R. Wang, Jundong Li. “Harnessing Large Language Models for Disaster Management: A Survey”, Findings of ACL (**ACL Findings**), 2025.
- [C9] Song Wang*, **Zhenyu Lei***, Zhen Tan, Jiaqi Ding, Xinyu Zhao, Yushun Dong, Guorong Wu, Tianlong Chen, Chen Chen, Aiyang Zhang, Jundong Li. “BrainMAP: Learning Multiple Activation Pathways in Brain Networks”, AAAI Conference on Artificial Intelligence (**AAAI**), 2025. (**Oral Presentation**)
- [C10] **Zhenyu Lei**, Yushun Dong, Jundong Li, Chen Chen. “ST-FiT: Inductive Spatial-Temporal Forecasting with Limited Training Data”, AAAI Conference on Artificial Intelligence (**AAAI**), 2025. (**Oral Presentation**) [Paper]
- [C11] **Zhenyu Lei***, Herun Wan*, Wenqian Zhang, Shangbin Feng, Jundong Li, Qinghua Zheng, Minnan Luo. “BIC: Twitter Bot Detection with Text-Graph Interaction and Semantic Consistency”, Annual Meeting of the Association for Computational Linguistics (**ACL**), 2023.
- [C12] Tianhao Huang, Guanghui Min, **Zhenyu Lei**, Aiyang Zhang, Chen Chen. “Uncovering Latent Communication Patterns in Brain Networks via Adaptive Flow Routing”, International Conference on Machine Learning (**ICML**), 2026.
- [C13] Xingbo Fu, **Zhenyu Lei**, Zihan Chen, Binchi Zhang, Chuxu Zhang, Jundong Li. “GraphTOP: Graph Topology-Oriented Prompting for Graph Neural Networks”, Annual Conference on Neural Information Processing Systems (**NeurIPS**), 2025.
- [C14] Zihan Chen, Song Wang, Xingbo Fu, Chengshuai Shi, **Zhenyu Lei**, Cong Shen, Jundong Li. “From Cross-Task Examples to In-Task Prompts: A Graph-Based Pseudo-Labeling Framework for In-context Learning”, Findings of EMNLP (**EMNLP Findings**), 2025.
- [C15] Yushun Dong, Song Wang, **Zhenyu Lei**, Zaiyi Zheng, Jing Ma, Chen Chen, Jundong Li. “Fairness-Aware Graph Learning: A Benchmark”, ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD**), 2025.
- [C16] Patrick Soga, **Zhenyu Lei**, Yinhan He, Camille L. Bilodeau, Jundong Li. “Energy-Based Models for Predicting Mutational Effects on Proteins”, ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD**), 2025.
- [C17] Patrick Soga, **Zhenyu Lei**, Camille L. Bilodeau, Jundong Li. “Deep Interactions for Multimodal Molecular Property Prediction”, Pacific-Asia Conference on Knowledge Discovery and Data Mining (**PAKDD**), 2025.
- [C18] Yushun Dong, Binchi Zhang, **Zhenyu Lei**, Na Zou, Jundong Li. “IDEA: A Flexible Framework of Certified Unlearning for Graph Neural Networks”, ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD**), 2024.
- [C19] Yushun Dong, **Zhenyu Lei**, Zaiyi Zheng, Song Wang, Jing Ma, Alex Jing Huang, Chen Chen, Jundong Li. “PyGDebias: A Python Library for Debiasing in Graph Learning”, Companion Proceedings of the ACM Web Conference (**WWW Companion**), 2024. [Code]
- [C20] Zijian Cai, Zhaoxuan Tan, **Zhenyu Lei**, Zifeng Zhu, Hongrui Wang, Qinghua Zheng, Minnan Luo. “LMBot: Distilling Graph Knowledge into Language Model for Graph-less Deployment in Twitter Bot Detection”, ACM International Conference on Web Search and Data Mining (**WSDM**), 2024.
- [C21] Shangbin Feng, Zhaoxuan Tan, Wenqian Zhang, **Zhenyu Lei**, Yulia Tsvetkov. “KALM: Knowledge-Aware Integration of Local, Document, and Global Contexts for Long Document Understanding”, Annual Meeting of the Association for Computational Linguistics (**ACL**), 2023.

- [C22] Shangbin Feng, Zhaoxuan Tan, Herun Wan, Ningnan Wang, Zilong Chen, Binchi Zhang, Qinghua Zheng, Wenqian Zhang, **Zhenyu Lei**, Shujie Yang, Xinshun Feng, Qingyue Zhang, Hongrui Wang, Yuhan Liu, Yuyang Bai, Heng Wang, Zijian Cai, Yanbo Wang, Lijing Zheng, Zihan Ma, Jundong Li, Minnan Luo. “Twibot-22: Towards Graph-Based Twitter Bot Detection”, NeurIPS Datasets and Benchmarks Track (**NeurIPS**), 2022.
- [C23] Wenqian Zhang, Shangbin Feng, Zilong Chen, **Zhenyu Lei**, Jundong Li, Minnan Luo. “KCD: Knowledge Walks and Textual Cues Enhanced Political Perspective Detection in News Media”, Annual Conference of the North American Chapter of the ACL (**NAACL**), 2022.

Journal Publications

- [J1] Zihan Chen, Song Wang, Zhen Tan, Xingbo Fu, **Zhenyu Lei**, Peng Wang, Huan Liu, Cong Shen, Jundong Li. “A Survey of Scaling in Large Language Model Reasoning”, ACM SIGKDD Explorations Newsletter (**SIGKDD Explorations**), 2026.
- [J2] Rong Ding, Weiyu Li, **Zhenyu Lei**, Yushun Dong, Jundong Li, Qi R. Wang. “Spatiotemporal Modeling of Social Capital and Infrastructure Mobility Resilience during Hurricane Harvey”, **Natural Hazards Review**, 2026.

Tutorials

- [T1] Xingbo Fu, Zehong Wang, Zihan Chen, Jiazheng Li, Yaochen Zhu, **Zhenyu Lei**, Cong Shen, Yanfang Ye, Chuxu Zhang, Jundong Li. “Graph Prompting for Graph Learning Models: Recent Advances and Future Directions”, ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD**), 2025.

PRESENTATIONS

- “MolEdit: Knowledge Editing for Multimodal Molecule Language Models” — Oral, WSDM 2026
- “ST-FiT: Inductive Spatial-Temporal Forecasting with Limited Training Data” — Oral, AAAI 2025
- “BrainMAP: Learning Multiple Activation Pathways in Brain Networks” — Oral, AAAI 2025

PROFESSIONAL SERVICES

Program Committee / Reviewer:

- Neural Information Processing Systems (NeurIPS), from 2023
- International Conference on Learning Representations (ICLR), 2025, 2026
- International Conference on Machine Learning (ICML), 2026
- Conference on Language Modeling (COLM), from 2024
- ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), from 2024
- ACM International Conference on Information and Knowledge Management (CIKM), 2025
- ACL Rolling Review, from December 2023

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